SAE Study — Findings

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All results are HuBERT on CREMA-D (lexicon-controlled: 91 actors × 12 fixed sentences × 6 emotions, 7,442 clips) unless noted. SAEs are TopK (d_sae=8192, k=32) trained on **per-frame** L-layer activations. Reconstruction is healthy: **FVU 0.070 (93% variance explained), 4/8192 dead features** (Step 1).

Method note — confound control was essential
Selecting "emotion features" by emotion-MI + monosemanticity alone flagged **394/8192** features, but the Step-3 control showed **66% of them carried more information about the spoken *sentence* (lexical/phonetic) or *speaker* than about emotion** — the documented "SAE rediscovers phonemes, not affect" failure mode. Requiring `MI(emotion) > MI(sentence)` and `> MI(actor)` on CREMA-D's factorial yields **61 genuine emotion features** (median monosemanticity 0.59), dominated by **anger (29) and sadness (18)**; fear/neutral drop to 0 — consistent with acoustic encoding of high-arousal, acoustically-marked emotions.

Q (i) — Depth: emotion *disentangles* earlier than it *decodes*
`step4_HuBERT_CREMA-D_depth.{csv,png}`

quantity	peak layer	value
linear probe accuracy (decodability)	**L10**	0.739
# monosemantic emotion features (disentanglement)	**L7**	113
monosemanticity of significant features	**L2**	541 sig-mono

The decode and disentanglement peaks **do not coincide**. Deep layers (L17–24) retain decodability (~0.72) but lose disentanglement (24–47 emotion features) — emotion stays linearly readable while becoming distributed/entangled. **Emotion organizes into clean monosemantic features in early-middle layers (L2–L7), before the layer where it is best linearly decoded (L10).**

Q (ii) — Generalization
ii.1 Across pretraining paradigms (`step5_*`): all 6 encoders carry healthy emotion-feature populations at their best layers (HuBERT 61, wav2vec2 70, Whisper 79, WavLM 145, MERT 64, w2v-BERT 61), all anger/sadness-dominated. Cross-encoder feature overlap (per-utterance firing match): **Whisper→SSL 0.40 ≈ SSL→SSL 0.39**.
→ On fixed-lexicon CREMA-D **Whisper is not distinct** from SSL in acoustic emotion features; its known divergence is about resisting the *lexical* shortcut, not about acoustic emotion encoding (which all encoders share). The lexical-divergence test is the deferred EMIS check (`STEP8_EMIS_STUB.md`).

ii.3 Across speakers (`step6_*`): the confound-controlled emotion features are **speaker-invariant** — random-CV 0.551 vs speaker-disjoint (GroupKFold over 91 actors) 0.551, **leave-speaker-out drop ≈ 0**, median actor coverage 0.95. Because selection excluded speaker-entangled features (`MI(emotion) > MI(actor)`), the published **MESD leave-one-speaker-out collapse (~0.30) is attributable to exactly those speaker-entangled features** the SAE lets us isolate.

ii.2 Across languages: BLOCKED — SpeechCraft audio is not on this cluster (`STEP6_CROSSLINGUAL_STUB.md`).

Q (iii) — Acoustic vs lexical: the signal is **acoustic**
Positive control (`step3_*`, fixed lexicon ⇒ must be acoustic): **60/61** emotion features label acoustic — mean eGeMAPS firing-AUC **0.83** vs sentence firing-AUC **0.66** (margin +0.17). The labeling method works.

Causal sufficiency (`step7_*`): emotion decodes from the 61 **acoustic** features at **0.552**, vs **0.431** from 61 lexical features and **0.423** from 61 random (6-class; chance ≈0.17, full reconstruction 0.73). Acoustic features carry **+0.12** more emotion than lexical → **the recovered emotion signal is acoustic, not lexical.** (Necessity-ablation of 61/8192 features is ~null because the signal is distributed; sufficiency is the informative test.)

Deferred (data not on cluster)
EMIS audio-vs-text feature separation (Step 8), IEMOCAP real-transcript ablation (Step 7 exact variant), SpeechCraft cross-lingual (Step 6) — each has a runnable `STEP*_STUB.md` with the precise procedure to run when the data is available.

SAE dissection of frozen-encoder emotion features (HuBERT / CREMA-D)